



اُنِيْوَرْسِيْتِيْ تِكْنُوْلُوْجِيْ مَآرَا  
UNIVERSITI  
TEKNOLOGI  
MARA

ITT270

DIGITAL ELECTRONICS

TITLE: SMART RFID AUTOMATED GATE SYSTEM

CLASS: M3CS1104C

PREPARED BY:

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## 1.0 MEMBERS PROFILES

### 1.1 ADAM ISKANDAR BIN NAZRI



# HI, I'M ADAM ISKANDAR BIN NAZRI

**ABOUT ME**

**Date of Birth:** 16/03/2006  
**Age:** 20  
**Hometown:** Paya Rumput, Melaka  
**Course:** CDCS110  
**Hobbies:** Playing basketball, Cooking  
**Student ID:** 2024277906  
**Strength:** Time management  
**Weakness:** Quiet and shy

**COMPUTER SKILLS**

Language: C++, Java, SQL, PHP  
Software: Oracle VirtualBox, BlueJ, xampp, Codeblocks, Wireshark

- Using Microsoft Office tools
- File management
- Basic troubleshooting
- Online collaboration tools

**EDUCATION BACKGROUND**

Primary School: SK Paya Rumput  
Secondary School: SMK Paya Rumput  
SPM Result: 8A  
Current CGPA: 3.76

**NEXT 5 YEARS PLANNING**

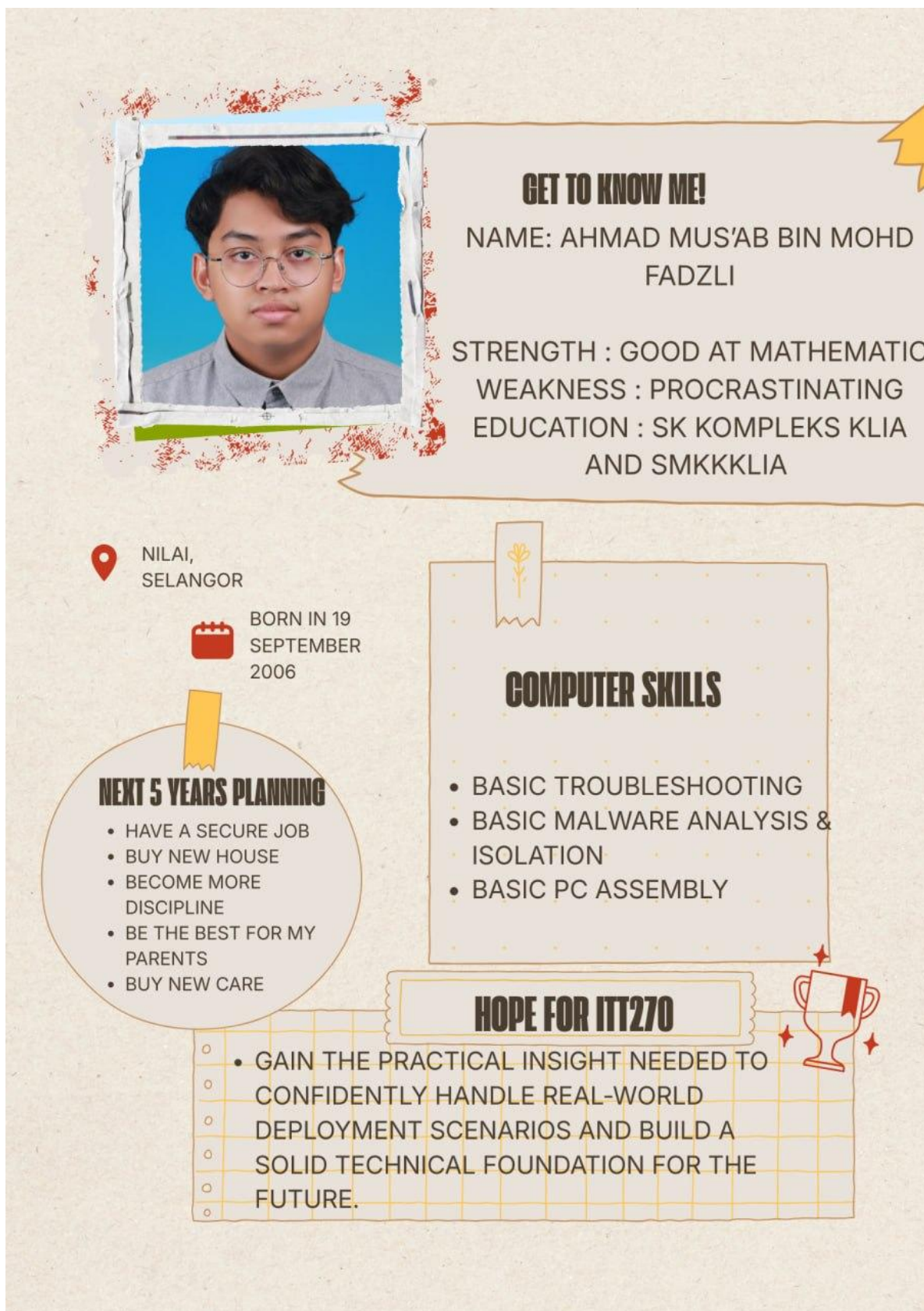
- Gain experience through internship
- Continue learning new skills
- Build discipline
- Secure a stable job
- Buy a car and a house
- Give back to my parents

**HOPE FOR ITT270**

- To improve my problem-solving and coding skills
- To complete all assignments and projects successfully
- Achieve a good grade in ITT270



## 1.2 AHMAD MUS'AB BIN MOHD FADZLI



The infographic is designed to look like a scrapbook or a collage of notes. It features a central photo of a young man with glasses, surrounded by various text boxes and decorative elements like torn paper edges, a location pin, a calendar icon, a flower clip, and a trophy icon. The background is a light beige color with a subtle grid pattern.

**GET TO KNOW ME!**

NAME: AHMAD MUS'AB BIN MOHD FADZLI

STRENGTH : GOOD AT MATHEMATIC  
WEAKNESS : PROCRASTINATING  
EDUCATION : SK KOMPLEKS KLIA AND SMKKKLIA

**NILAI, SELANGOR**

**BORN IN 19 SEPTEMBER 2006**

**NEXT 5 YEARS PLANNING**

- HAVE A SECURE JOB
- BUY NEW HOUSE
- BECOME MORE DISCIPLINE
- BE THE BEST FOR MY PARENTS
- BUY NEW CAR

**COMPUTER SKILLS**

- BASIC TROUBLESHOOTING
- BASIC MALWARE ANALYSIS & ISOLATION
- BASIC PC ASSEMBLY

**HOPE FOR ITT270**

- GAIN THE PRACTICAL INSIGHT NEEDED TO CONFIDENTLY HANDLE REAL-WORLD DEPLOYMENT SCENARIOS AND BUILD A SOLID TECHNICAL FOUNDATION FOR THE FUTURE.

## 1.3 MUHAMMAD FAKRUL BIN ABU KASIM



### MUHAMMAD FAKRUL BIN ABU KASIM

#### ABOUT ME

STUDENT ID : 2024413192  
DATE OF BIRTH : 5 / 9 / 2006  
AGE : 20  
HOMETOWN : PAYA RUMPUT, MELAKA  
COURSE : CDCS110  
HOBBIES : PLAYING FOOTBALL

#### EDUCATION



##### PRIMARY SCHOOL

2013 – 2018  
SK PAYA RUMPUT



##### SECONDARY SCHOOL

2019 – 2023  
SMK PAYA RUMPUT



##### SPM RESULT

5A



##### CURRENT CGPA RESULT

3.82

#### SKILLS

- Language : C++, Java , SQL
- Software : VirtualBox, BlueJ , Visual Studio Code, Adobe Photoshop
- Microsoft Office
- Canva
- Communication
- Teamwork
- Leadership

#### NEXT 5 YEAR PLANNING

- GRADUATE WITH EXCELLENT ACADEMIC PERFORMANCE
- BEGIN A CAREER IN MY CHOSEN FIELD
- DEVELOP STRONG TECHNICAL AND SOFT SKILLS
- GAIN EXPERIENCE THROUGH PROJECTS AND INDUSTRY EXPOSURE
- PROGRESS TOWARDS HIGHER ROLES AND RESPONSIBILITIES
- IMPROVE MY PROBLEM SOLVING AND CODING SKILLS
- ACHIEVE GOOD RESULT IN ITT270
- COMPLETE ALL ASSIGNMENT AND GET GOOD MARK

## 1.4 MUHAMMAD THAQIF BIN MOHD ZULFATAH

# MUHAMMAD THAQIF BIN MOHD ZULFATAH

 STUDENT ID: 2024268916



### ABOUT ME

-  **Date of Birth:** 03/02/2006
-  **Age:** 20
-  **Hometown:** Merlimau, Melaka
-  **Course:** CDCS110
-  **Hobbies:** Running and playing football
-  **Strength:** Time management
-  **Weakness:** Quiet and shy

### COMPUTER SKILLS

-  **Programming Languages:**  
C++, Java, SQL, PHP
-  **Software:**  
Oracle VirtualBox, BlueJ, XAMPP, Codeblocks, Wireshark
-  Using Microsoft Office tools
-  File management
-  Basic troubleshooting
-  Online collaboration tools

### EDUCATION BACKGROUND

- Primary School:** SK KGV
- Secondary School:** SMK KING GEORGE V
- SPM Result:** 8A
- Current CGPA:** 3.7



### NEXT 5 YEARS PLANNING

- Gain experience through internship
- Continue learning new skills
- Build discipline
- Secure a stable job
- Buy a car and a house
- Give back to my parents



### HOPE FOR ITT270

- To deeply understand networking fundamentals and data communication concepts.
- To actively apply theoretical knowledge during hands-on practical assignments.
- To achieve an outstanding grade and enhance my technical problem-solving abilities.





## 1.5 MUHAMMAD THAQIF BIN AZNAM



A portrait of a young man with dark hair and a slight beard, wearing a patterned brown shirt. The photo is framed by a black border and decorated with white clouds and a yellow lightning bolt on the left.

# MUHAMAD THAQIF

### ABOUT ME

- NAME : MUHAMAD THAQIF BIN AZNAM
- ID STUDENT : 2024694282
- BIRTH DATE : 18/01/2006
- COURSE : COMPUTER SCIENCE
- HOBBY : PLAYING VOLLEYBALL



### EDUCATION BACKGROUND

- **Universiti Teknologi MARA (UiTM), Kampus Jasin**  
Diploma in Computer Science  
2023 - Present
- **SMK Seksyen 10 Kota Damansara**  
Secondary Education (Form 1 - Form 5)  
2018 - 2023
- **Nichols Junior High School**  
Middle School Education  
2017 - 2018
- **Pakistan National English School**  
Primary Education  
2016 - 2017
- **As-Sofa Islamic School**  
Primary Education  
2013 - 2016
- **PASTI Little Mukmin**  
Preschool Education  
2010 - 2012

### 5 YEARS PLANNING

- Graduate with First Class.
- Study Degree in Artificial Intelligence.
- Go to Makkah for Umrah.
- Visit Morocco.
- Start my own tech project or small business

### TARGET FOR ITT 270

- **Achieve Excellence:** Strive to obtain an A grade through consistent effort, discipline, and smart learning strategies.
- **Deepen Understanding:** Gain a comprehensive grasp of the subject's core concepts and real-world applications.
- **Enhance Technical Proficiency:** Strengthen practical IT and problem-solving skills to apply knowledge effectively in future coursework.
- **Develop Critical Thinking:** Cultivate a mindset that encourages innovation, analysis, and creativity in tackling ITT 270 challenges.



### COMPUTER SKILL

- **Programming Languages :** Basic proficiency in Java and C++; beginner experience with Python (learned in high school).
- **Video Editing:** Skilled in Adobe Animate and Canva for creating and editing videos and animations.
- **Graphic Design:** Familiar with Adobe Photoshop for image editing and basic design tasks.
- **General Skills:** Strong understanding of computer operations, file management, and creative digital tools.



## 2.0 INTRODUCTION

The rapid evolution of smart home technology has transformed residential security, moving properties away from the vulnerabilities of physical keys and standard remote controls. To address the security risks and access delays associated with legacy systems, this project presents the **Smart RFID & Keypad Automated Gate System with Night-Light Integration**. By merging hardware automation with multi-factor authentication, this prototype bridges the gap between robust perimeter security and everyday user convenience.

At its core, the project utilizes a dual-authentication framework. Primary access is regulated through a contactless RFID subsystem for residents, while a secondary 4x4 matrix keypad provides a secure, four-digit PIN override for trusted guests and family members. To optimize power and safety, ultrasonic sensor arrays dynamically monitor approaching vehicles, keeping the system in low-power standby until a car arrives.

Beyond security, the system incorporates intelligent environmental awareness. A built-in Photoresistor (LDR) continuously gauges ambient light levels, automatically illuminating a driveway courtesy light when a vehicle approaches during nocturnal hours. Governed by a central microcontroller and providing real-time visual feedback via an LCD screen, this project successfully demonstrates a secure, energy-efficient, and scalable blueprint for modern IoT residential access.



### 3.0 PROBLEM STATEMENT

Modern residential perimeters frequently suffer from security gaps and energy inefficiencies due to rigid, non-automated entry systems. For instance, relying strictly on physical access keys or remote fobs creates a single point of failure if they are lost or forgotten, leaving residents locked out and properties exposed. At the same time, legacy systems offer no secure, independent access methods for expected visitors, couriers, or emergency services, causing unnecessary entry delays.

Moreover, conventional outdoor security lighting operates on simple, inefficient timers or remains manual. This results in dark, hazardous driveways when vehicles arrive at night, or unneeded energy consumption if lights are left running during bright daylight hours.

To overcome these issues, a smart, automated gate entry system featuring dual-factor authentication and responsive, energy-saving environmental lighting is required.



| Car At Gate | Authorized RFID / PIN | Ambient Light (LDR) | Gate Servo | Driveway Light |
|-------------|-----------------------|---------------------|------------|----------------|
| Yes         | Yes                   | Dark (Night)        | OPEN       | ON             |
| Yes         | Yes                   | Bright (Day)        | OPEN       | OFF            |
| Yes         | No                    | —                   | LOCKED     | OFF            |
| No          | —                     | —                   | LOCKED     | OFF            |

## 4.0 TRUTH TABLE

Inputs:

| Variable                  | 1 = True Condition | 0 = False Condition     |
|---------------------------|--------------------|-------------------------|
| <b>CAR</b> (Ultrasonic)   | 1 = Car Present    | 0 = No Car              |
| <b>AUTH</b> (RFID / PIN)  | 1 = Authorized     | 0 = Unauthorized / None |
| <b>LDR</b> (Light Sensor) | 1 = Dark (Night)   | 0 = Bright (Day)        |

Outputs:

| Variable                    | 1 = Active State | 0 = Inactive State |
|-----------------------------|------------------|--------------------|
| <b>GATE</b> (Servo)         | 1 = Open         | 0 = Locked         |
| <b>LIGHT</b> (Driveway LED) | 1 = On           | 0 = Off            |

| INPUT   |          |         | OUTPUT |       |
|---------|----------|---------|--------|-------|
| CAR (A) | AUTH (B) | LDR (C) | GATE   | LIGHT |
| 0       | 0        | 0       | 0      | 0     |
| 0       | 0        | 1       | 0      | 0     |
| 0       | 1        | 0       | 0      | 0     |
| 0       | 1        | 1       | 0      | 0     |
| 1       | 0        | 0       | 0      | 0     |
| 1       | 0        | 1       | 0      | 1     |
| 1       | 1        | 0       | 1      | 0     |
| 1       | 1        | 1       | 1      | 1     |

## 4.1 KARNAUGH MAP

### 1. K-Map for GATE (Servo)

The gate only opens if a car is present (A) AND the user is authorized (B), regardless of whether it is day or night (C).

|    | A'B' | A'B | AB | AB' |
|----|------|-----|----|-----|
| C' | 0    | 0   | 1  | 0   |
| C  | 0    | 0   | 1  | 0   |

**Grouping:** A vertical pair in the **AB** column. (The C variable drops out).

**Equation:**  $\text{GATE} = AB$

### 2. K-Map for LIGHT (Driveway LED)

The driveway light only turns on if a car is present (A) AND it is dark out (C), regardless of whether they have scanned a valid tag yet (B).

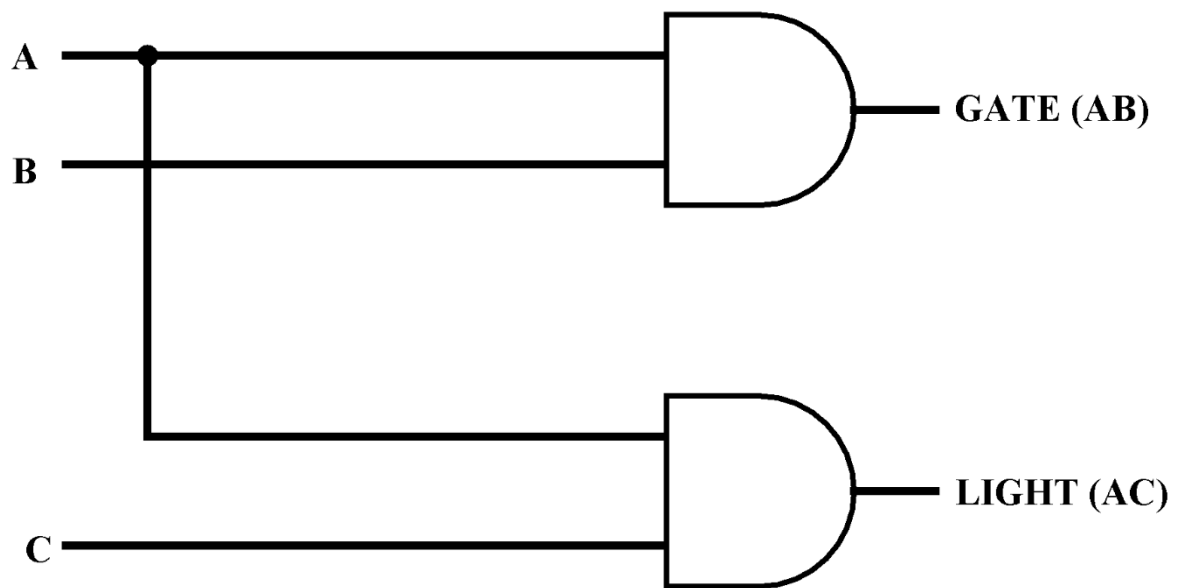
|    | A'B' | A'B | AB | AB' |
|----|------|-----|----|-----|
| C' | 0    | 0   | 0  | 0   |
| C  | 0    | 0   | 1  | 1   |

**Grouping:** A horizontal pair in the bottom row (C) spanning across columns **AB** and **AB'**. (The B variable drops out).

**Equation:**  $\text{LIGHT} = AC$



## 4.2 CIRCUIT MAKER



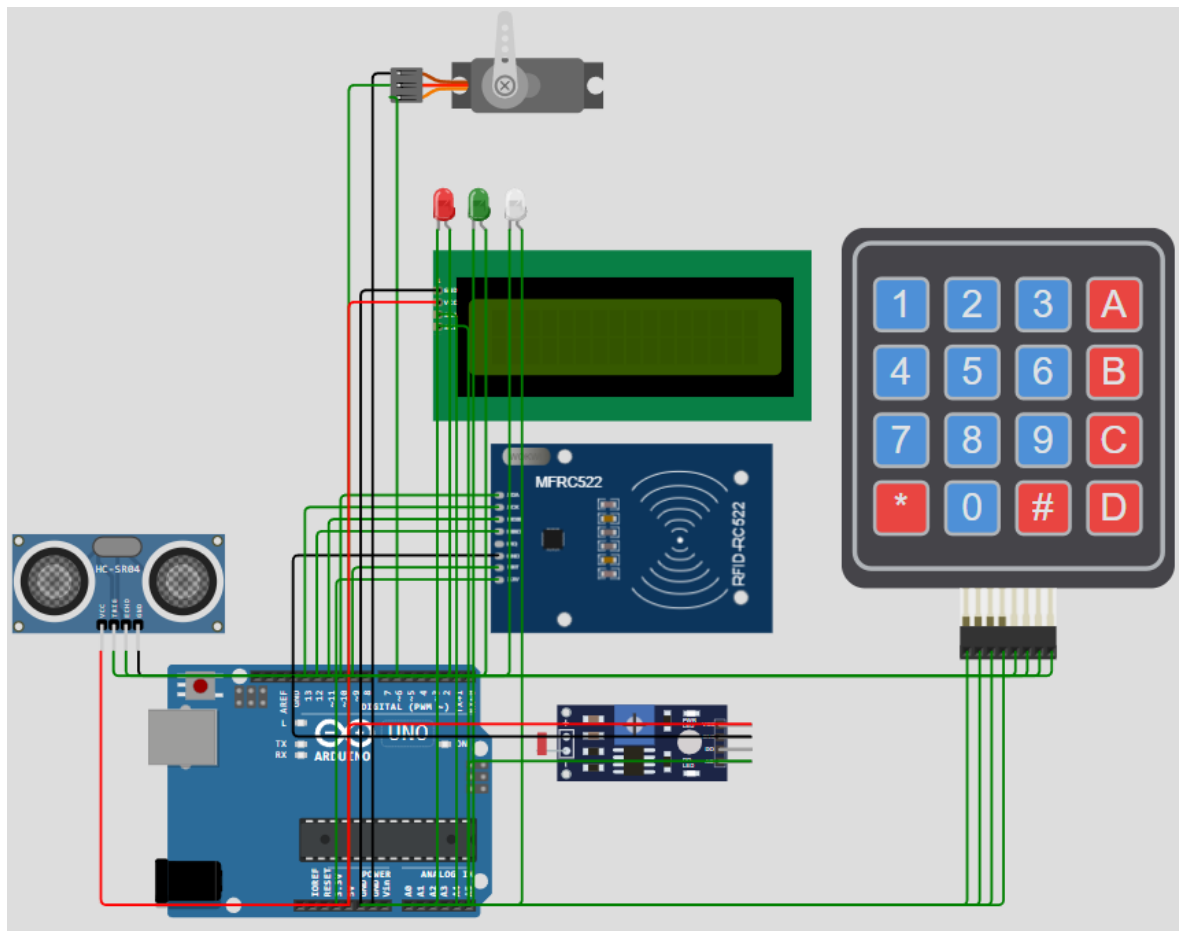
## 5.0 INTERNET OF THINGS (IoT)

### 5.1 IoT COMPONENTS

The Smart RFID Automated Gate System integrates several hardware components to achieve its dual-authentication and energy-saving capabilities. The system is governed by an **Arduino Uno** microcontroller, which processes inputs from sensors and manages the output actuators.

- **Arduino Uno:** Acts as the central processing unit, executing the control logic for the gate and lighting subsystems.
- **MFRC522 RFID Reader:** Facilitates the primary resident authentication by scanning contactless RFID tags.
- **HC-SR04 Ultrasonic Sensor:** Measures the proximity of approaching vehicles to trigger the system's "Car Present" logic.
- **4x4 Membrane Keypad:** Provides the secondary authentication method, allowing authorized guests or family members to enter via a secure 4-digit PIN.
- **Photoresistor (LDR):** Monitors ambient light levels to determine if it is "Night" (Dark) or "Day" (Bright), enabling automated energy-saving light control.
- **Servo Motor:** Functions as the gate actuator, opening the gate when the system validates authorized access.
- **LCD 1602 (I2C):** Displays real-time status updates and provides visual feedback to the user regarding the gate's status.
- **LED Indicators:**
  - **Red LED:** Provides visual feedback for denied or unsuccessful access attempts.
  - **Green LED:** Indicates successful authentication and that the gate is opening.
  - **White LED:** Serves as the automated driveway light that illuminates based on environmental conditions.

## 5.2 IoT DIAGRAM



## 5.3 IoT CODING

```
1  #include <SPI.h>
2  #include <MFRC522.h>
3  #include <Servo.h>
4  #include <Wire.h>
5  #include <LiquidCrystal_I2C.h>
6  #include <Keypad.h>
7
8  #define RST_PIN      9
9  #define SS_PIN      10
10 #define TRIG_PIN     2
11 #define ECHO_PIN     3
12 #define GREEN_LED    7
13 #define RED_LED      8
14 #define DRIVEWAY_LIGHT A4      // White LED shared on I2C line cleanly
15 #define LDR_PIN      A5      // LDR Analog Input
16
17 const byte ROWS = 4;
18 const byte COLS = 4;
19 char hexaKeys[ROWS][COLS] = {
20     {'1','2','3','A'},
21     {'4','5','6','B'},
22     {'7','8','9','C'},
23     {'*','0','#','D'}
24 };
25 byte rowPins[ROWS] = {A0, A1, A2, A3};
26 byte colPins[COLS] = {0, 1, 4, 5};
27 Keypad customKeypad = Keypad(makeKeymap(hexaKeys), rowPins, colPins, ROWS, COLS);
28
29 MFRC522 rfid(SS_PIN, RST_PIN);
30 Servo gateServo;
31 LiquidCrystal_I2C lcd(0x27, 16, 2);
32
33 byte ownerUID[4] = {0x11, 0x22, 0x33, 0x44};
34 const String secretPIN = "1234";
35 String inputPIN = "";
36
37 void setup() {
38     SPI.begin();
39     rfid.PCD_Init();
40     gateServo.attach(6);
41
42     pinMode(TRIG_PIN, OUTPUT);
43     pinMode(ECHO_PIN, INPUT);
44     pinMode(GREEN_LED, OUTPUT);
45     pinMode(RED_LED, OUTPUT);
46     pinMode(DRIVEWAY_LIGHT, OUTPUT);
47
48     lcd.init();
49     lcd.backlight();
50     resetSystemView();
51 }
52
53 void loop() {
54     long distance = getDistance(TRIG_PIN, ECHO_PIN);
55     char customKey = customKeypad.getKey();
56
57     int lightLevel = analogRead(LDR_PIN);
58     bool isDark = (lightLevel > 600);
59
60     if (distance < 50) {
61         lcd.setCursor(0, 0);
62         lcd.print("Car At Gate!   ");
63
64         if (isDark) {
65             digitalWrite(DRIVEWAY_LIGHT, HIGH);
```



```

66     }
67
68     if (customKey) {
69         if (customKey == '#') {
70             if (inputPIN == secretPIN) {
71                 handleAccessGranted("VIP Guest");
72             } else {
73                 handleAccessDenied();
74             }
75             inputPIN = "";
76         } else if (customKey == '*') {
77             inputPIN = "";
78             lcd.setCursor(0, 1);
79             lcd.print("Cleared!      ");
80             delay(1000);
81         } else {
82             inputPIN += customKey;
83             lcd.setCursor(0, 1);
84             lcd.print("PIN: ");
85             for(int i=0; i<inputPIN.length(); i++) lcd.print("*");
86             lcd.print("      ");
87         }
88     } else if (inputPIN.length() == 0) {
89         lcd.setCursor(0, 1);
90         lcd.print("Scan RFID / PIN ");
91     }
92
93     if (rfid.PICC_IsNewCardPresent() && rfid.PICC_ReadCardSerial()) {
94         if (isOwnerCard()) {
95             handleAccessGranted("Owner");
96         } else {
97             handleAccessDenied();
98         }
99         rfid.PICC_HaltA();
100     }
101 } else {
102     resetSystemView();
103     digitalWrite(DRIVEWAY_LIGHT, LOW);
104     inputPIN = "";
105 }
106 delay(50);
107 }
108
109 long getDistance(int trig, int echo) {
110     digitalWrite(trig, LOW);
111     delayMicroseconds(2);
112     digitalWrite(trig, HIGH);
113     delayMicroseconds(10);
114     digitalWrite(trig, LOW);
115     long duration = pulseIn(echo, HIGH);
116     return duration * 0.034 / 2;
117 }
118
119 bool isOwnerCard() {
120     for (byte i = 0; i < 4; i++) {
121         if (rfid.uid.uidByte[i] != ownerUID[i]) return false;
122     }
123     return true;
124 }
125

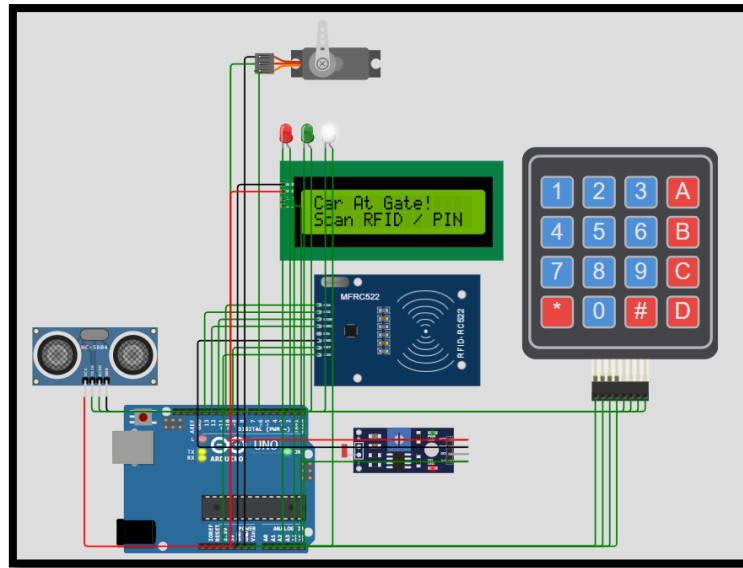
```

```

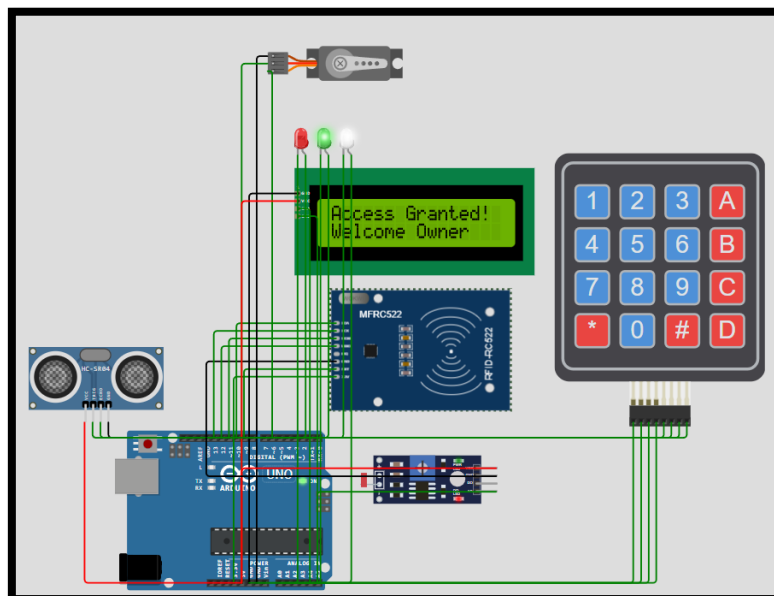
126 void handleAccessGranted(String userType) {
127     lcd.clear();
128     lcd.print("Access Granted!");
129     lcd.setCursor(0, 1);
130     lcd.print("Welcome " + userType);
131     digitalWrite(GREEN_LED, HIGH);
132
133     gateServo.write(90);
134     delay(5000);
135
136     lcd.clear();
137     lcd.print("Closing Gate... ");
138     gateServo.write(0);
139     digitalWrite(GREEN_LED, LOW);
140 }
141
142 void handleAccessDenied() {
143     lcd.clear();
144     lcd.print("INVALID ACCESS!");
145     lcd.setCursor(0, 1);
146     lcd.print("Gate Locked ");
147     digitalWrite(RED_LED, HIGH);
148     delay(3000);
149     digitalWrite(RED_LED, LOW);
150 }
151
152 void resetSystemView() {
153     lcd.setCursor(0, 0);
154     lcd.print("SECURE GATE SYS ");
155     lcd.setCursor(0, 1);
156     lcd.print("Status: Locked ");
157     gateServo.write(0);
158 }

```

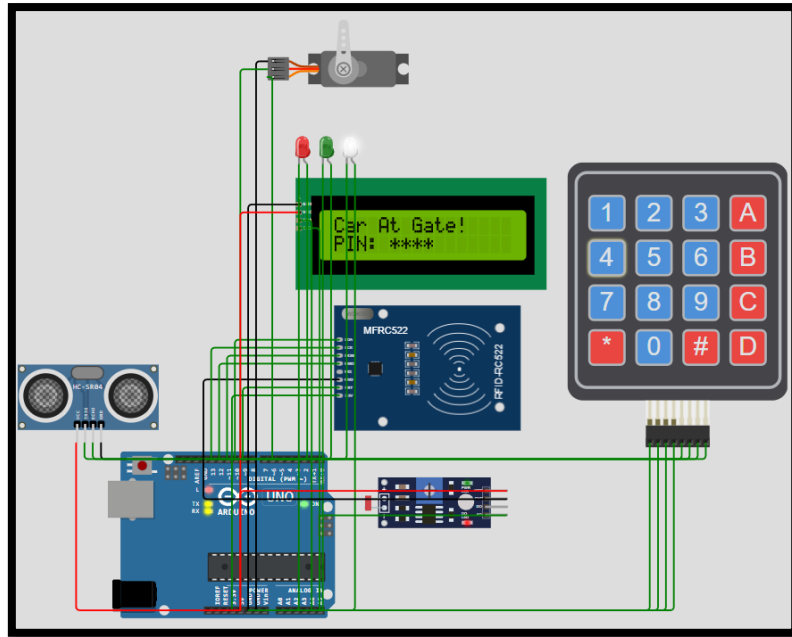
## 6.0 APPENDIX



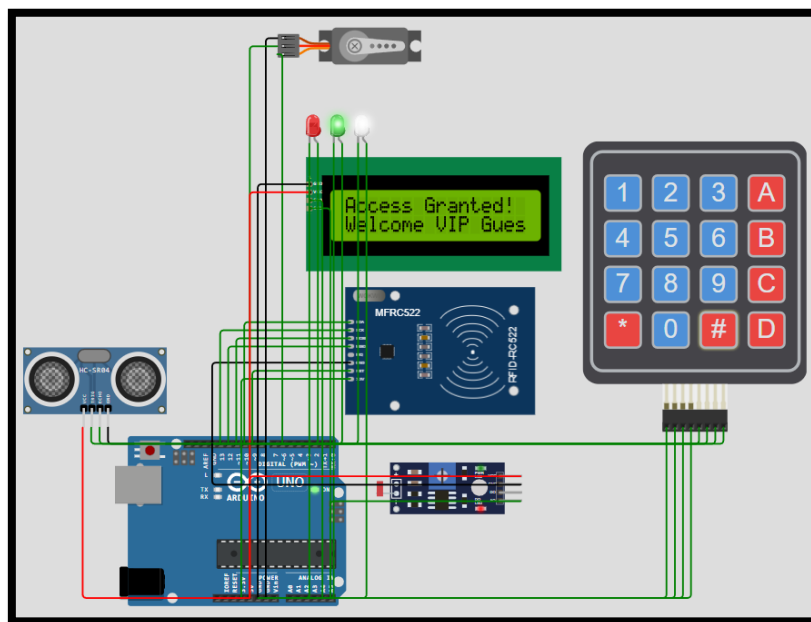
When a car is present at the gate.



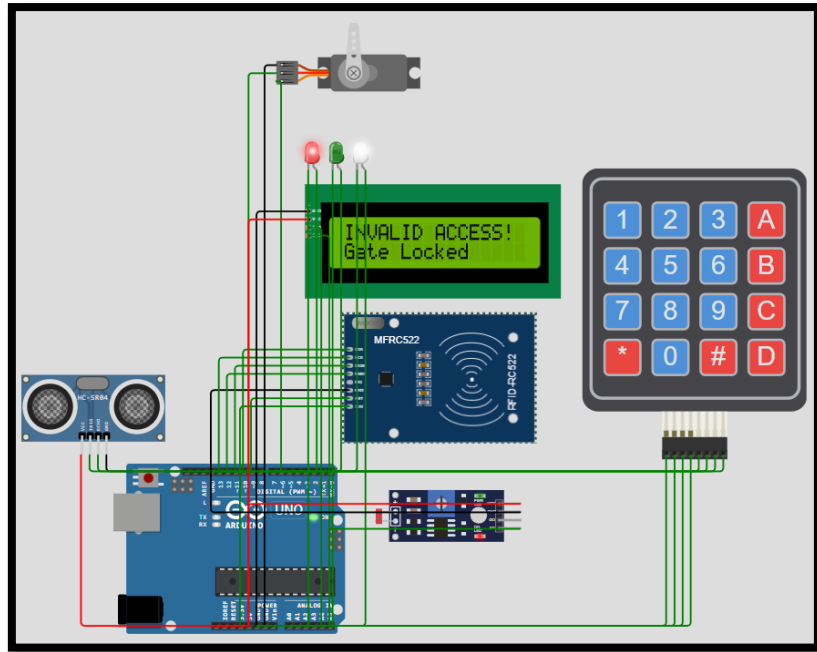
When authorized using the owner's card.



Inputting the pin (1234) for VIP access.



VIP access granted when inputting the right pin.



When using the wrong card or wrong input.



## 7.0 REFERENCES

1. Monk, S. (2022). *Programming Arduino: Getting Started with Sketches*. 3rd Edition. McGraw Hill.
2. Mano, M. M., & Ciletti, M. D. (2017). *Digital Design: With an Introduction to the Verilog HDL, VHDL, and SystemVerilog*. 6th Edition. Pearson.
3. Hazzah, M. (2023). "Design and Implementation of IoT-Based Security Systems Using RFID and Microcontrollers." *International Journal of Computer Applications*, 185(12), 34–39.
4. Arduino Documentation. (2026). *MFRC522 RFID Module Library and HC-SR04 Ultrasonic Sensor Reference*. Arduino.cc.
5. Smith, J. A. (2025). "Energy-Efficient Automation in Smart Home Environments." *Journal of Sustainable Engineering*, 9(4), 112–120.